EE 3072
System Dynamics and Control

Lecture 8
PID Controller Design

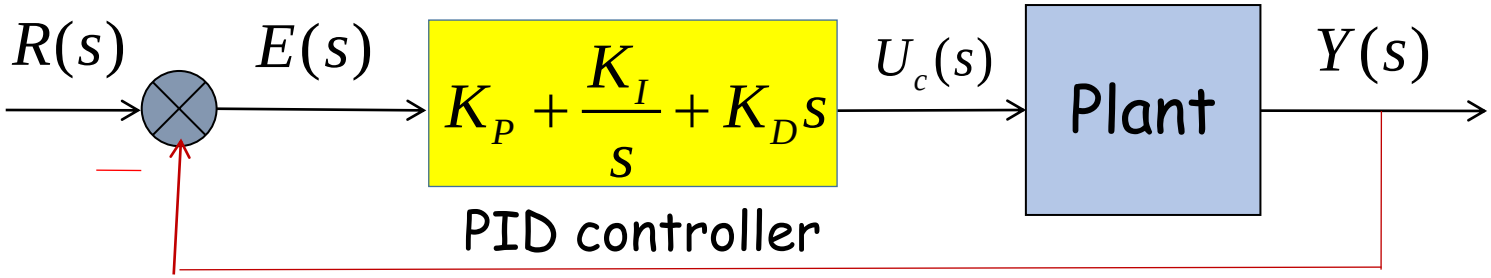
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PID Control

Proportional-Integral-Derivative (PID) Controller



TF of a PID controller

$$G_c(s) = \left(K_P + \frac{K_I}{s} + K_D s \right)$$

PID controller in time domain

$$u_c(t) = K_P e(t) + K_I \int_0^t e(\tau) d\tau + K_D \dot{e}(t)$$

Proportional (P) controller: $G_c(s) = K_P$

Proportional-Integral (PI) controller: $G_c(s) = K_P + \frac{K_I}{s}$

Proportional-Derivative (PD) controller: $G_c(s) = K_P + K_D s$

PID Control

PID controllers are effective controllers which are extensively used in practice.

More than half of the industrial controllers in use today utilize PID or modified PID control schemes.

When the mathematical model of the plant is unknown and therefore analytical design cannot be used, PID control proves to be useful.

Various types of tuning rules have been developed to adjust the parameters of PID controllers on-site.

P-Controller

Proportional (P) Controller TF: $G_c(s) = K_P$

In P-controllers we have K_I=0 and K_D=0, whereas $K_P \neq 0$.

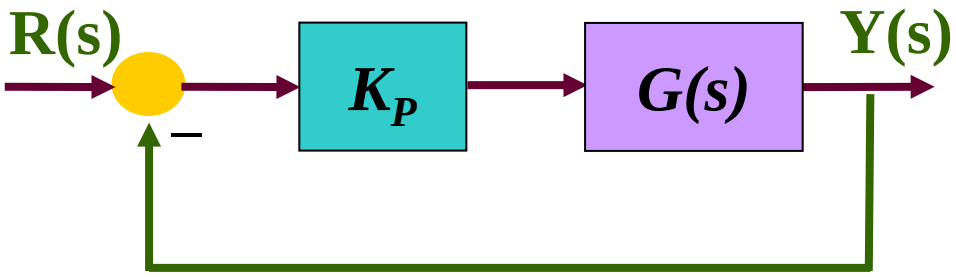
Note that P-controllers are not dynamic controllers.

Given desired performance specification the gain K_P of a P-controller can be chosen utilizing the root-locus method (discussed earlier).

Alternatively, one can also utilize the frequency response (the Nyquist plot and/or the Bode diagram).

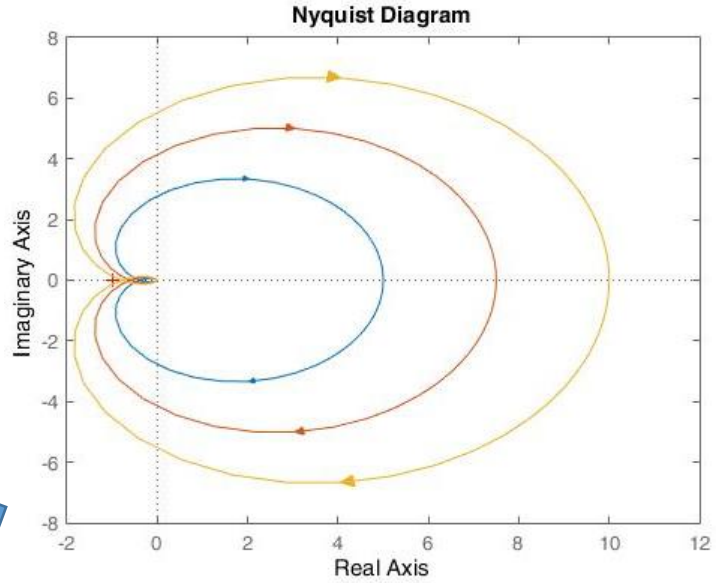
P-Controller

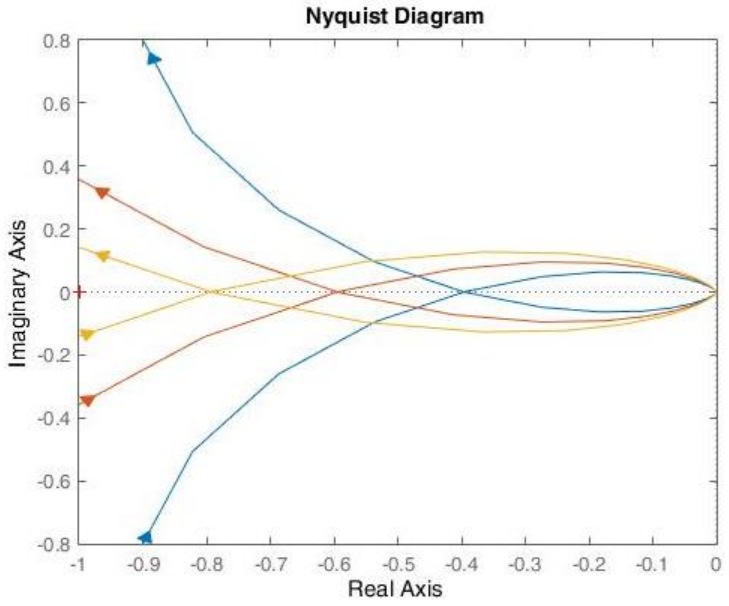
Consider the system



By varying the gain K_P one can enlarge or shrink the Nyquist plot.

However, by utilizing the gain K_P only, one cannot rotate or change the shape of the Nyquist plot.





P-Controller

The gain K_P changes the magnitude plot of the Bode diagram of the OL TF. However, its phase plot remains the same.

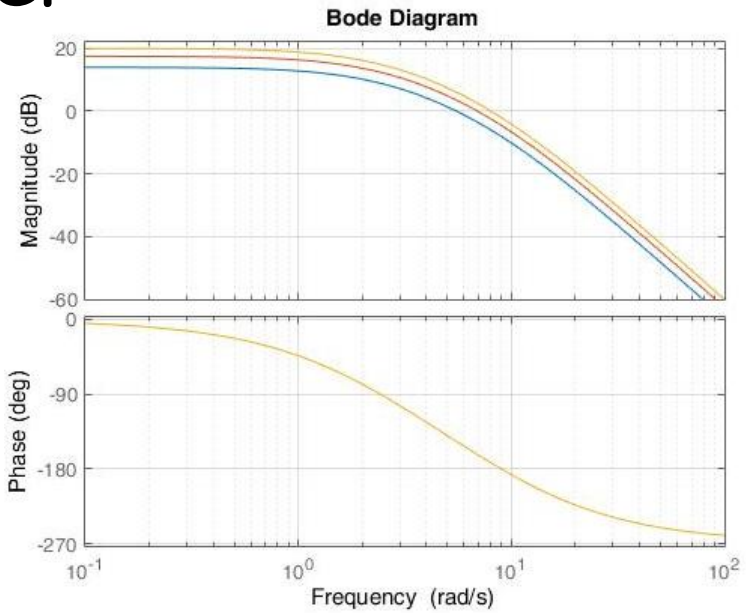
For the given Bode diagram

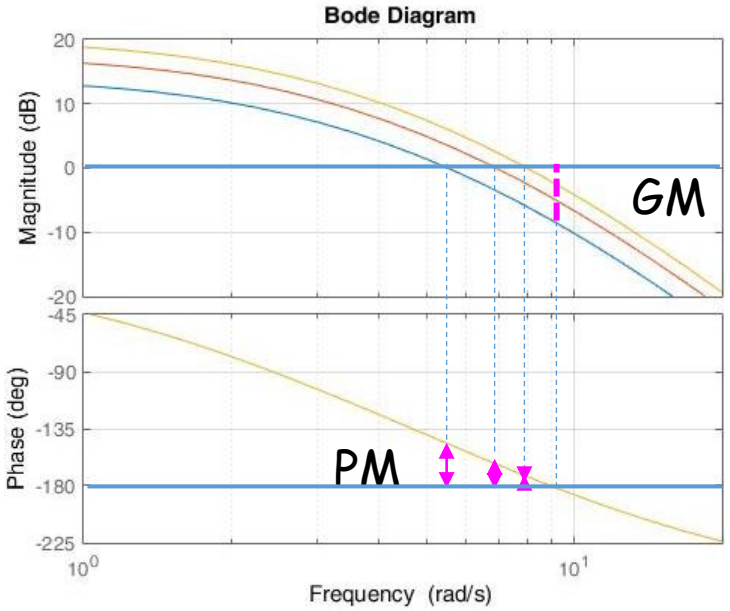
For K_P small (blue curve): The system is relatively more stable.

For K_P large (orange curve): The system is relatively less stable.

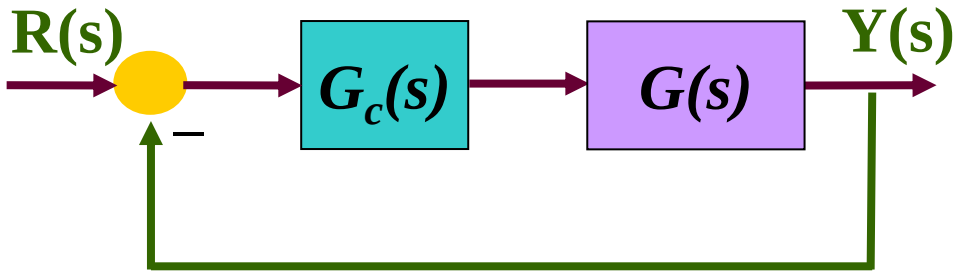
As K_P decreases, the phase and gain margins increase. However, the bandwidth decreases.

As K_P increases, the bandwidth increases. However, the phase and gain margins decrease.





PI-Controller



Proportional-Integral (PI) Controller TF:

$$G_c(s) = K_P + \frac{K_I}{s}$$

In PI-controllers we have K_D=0, whereas $K_P \neq 0$ and $K_I \neq 0$.

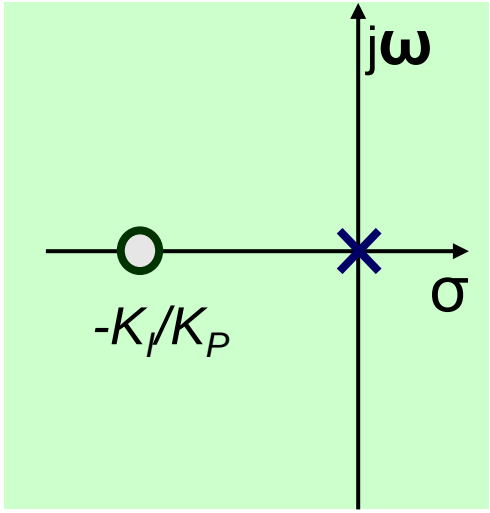
$$G_c(s) = K_P + \frac{K_I}{s} = \frac{K_P s + K_I}{s} = \frac{K_P (s + K_I / K_P)}{s}$$

$$zero: \quad s = -K_I / K_P, \qquad pole: \quad s = 0$$

Its sinusoidal TF is

$$G_c(j\omega) = \frac{K_I + j\omega K_P}{j\omega} = \frac{K_I (1 + j\omega K_P / K_I)}{j\omega}$$

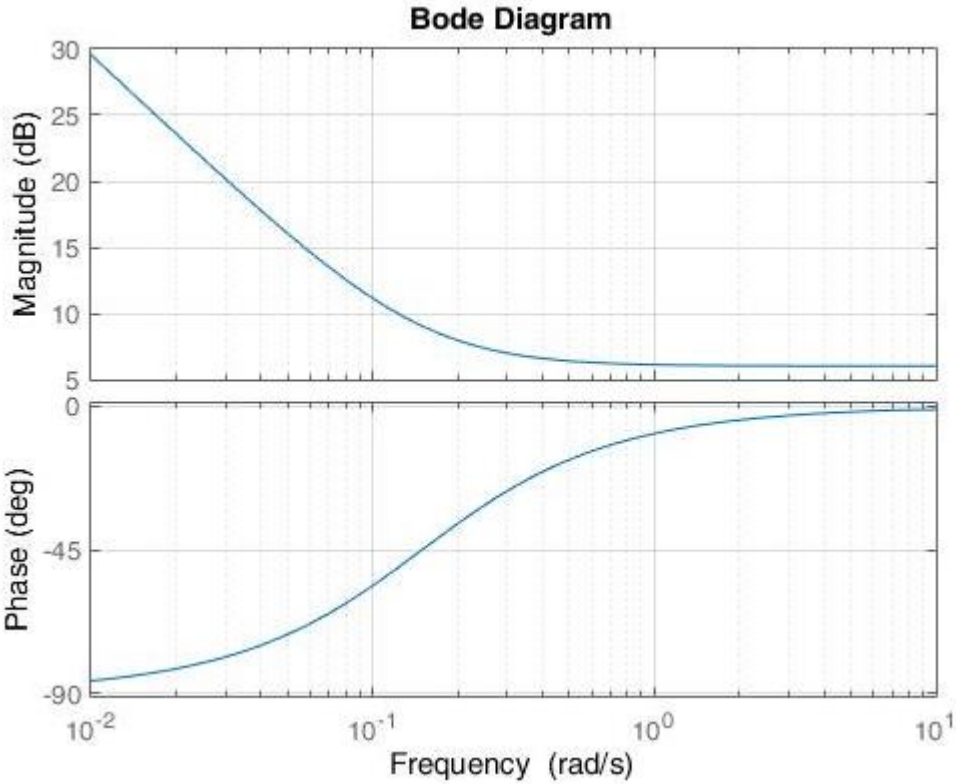
Pole-zero configuration of a PI-controller.



PI-Controller

$$G_c(j\omega) = \frac{K_I(1 + j\omega K_P / K_I)}{j\omega}$$

Its Bode diagram has the form



Bode diagram of a PI controller with K_P = 2 and K_I=0.3.

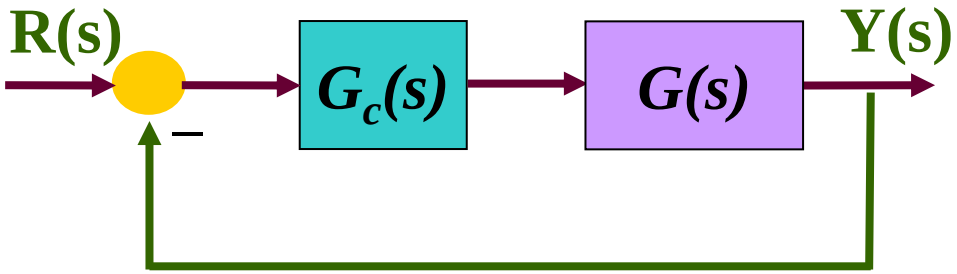
Note that a PI controller is a phase-lag controller.

It adds a negative phase (a phase-lag) to the system.

It increases the system type by one and is used to improve the steady-state accuracy of the system.

In order to determine the coefficients K_P and K_I, one can use a procedure similar to the procedure for a phase-lag controller design.

PD-Controller



Proportional-Derivative (PD) Controller TF:

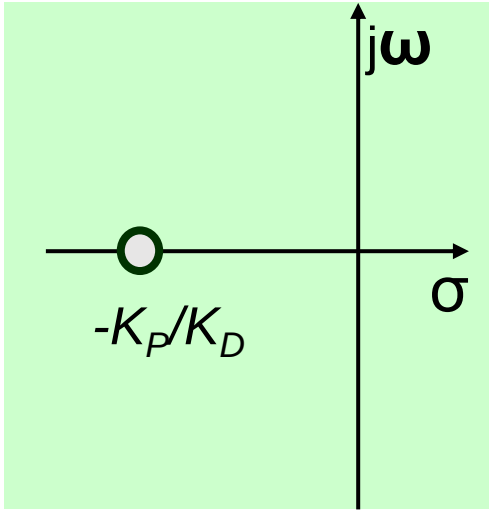
$$G_c(s) = K_P + K_D s$$

In PD-controllers we have K_I=0, whereas $K_P \neq 0$ and $K_D \neq 0$.

$$G_c(s) = K_P + K_D s = K_D(s + K_P / K_D)$$

$$zero: \quad s = -K_P / K_D, \quad pole: \quad s = -\infty$$

Pole-zero configuration of a PD-controller.



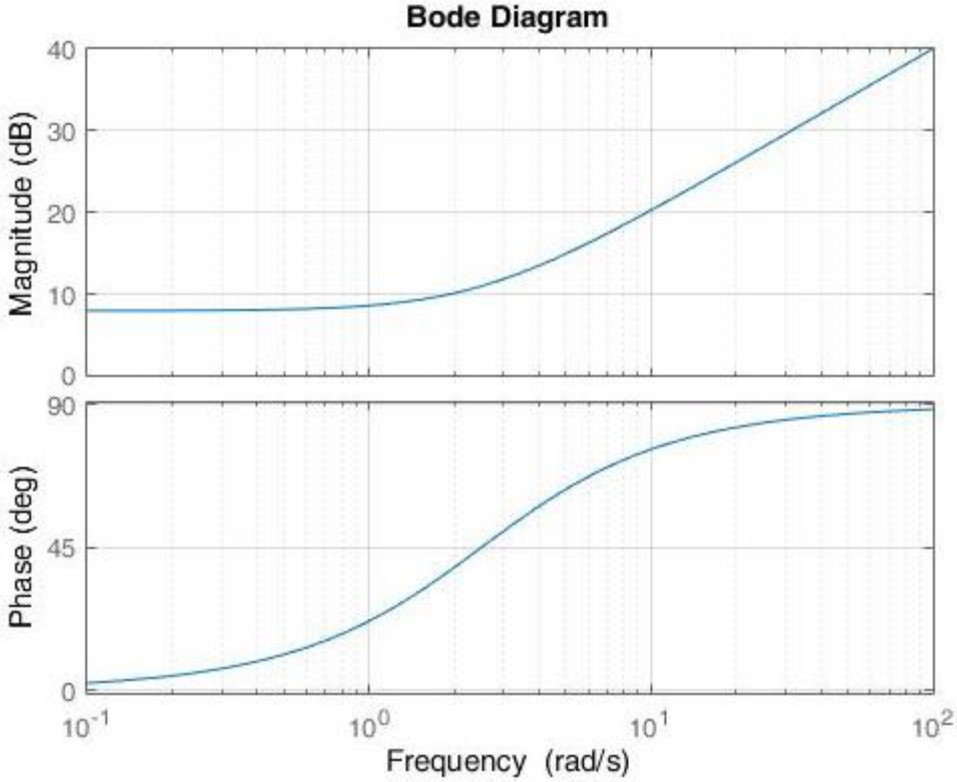
Its sinusoidal TF is

$$G_c(j\omega) = K_P + j\omega K_D = K_P(1 + j\omega K_D / K_P)$$

PD-Controller

$$G_c(j\omega) = K_P(1 + j\omega K_D / K_P)$$

Its Bode diagram has the form



Bode diagram of a PD controller with K_P = 1 and K_D=0.4.

Note that a PD controller is a phase-lead controller.

It adds a positive phase (a phase-lead) to the system.

It is used to improve the transient performance of the system.

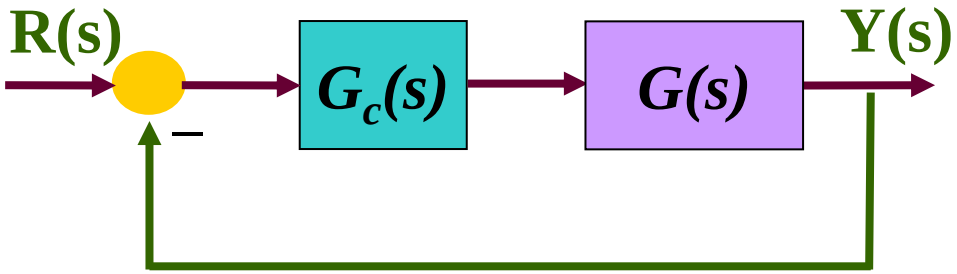
In order to determine the coefficients K_P and K_D, one can use a procedure similar to the procedure for a phase-lead controller design.

PD-Controller

One practical problem with PD-controllers is that the gain (i.e., the magnitude of the sinusoidal TF of the controller) continues to increase as the frequency increases.

Therefore, high frequency signals such as noise are amplified by PD controllers, which may have negative effects on the system response.

PID-Controller



Proportional-Integral-Derivative (PID) Controller TF:

$$G_c(s) = K_P + \frac{K_I}{s} + K_D s$$

When all $K_P \neq 0$, $K_I \neq 0$, and $K_D \neq 0$, then we have a PID controller.

$$G_c(s) = \frac{K_D s^2 + K_P s + K_I}{s}$$

$$= \frac{K_D\left(s^2 + (K_P / K_D)s + K_I / K_D\right)}{s}$$

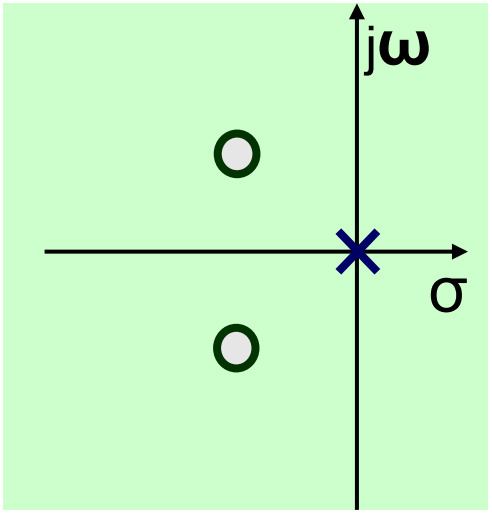
$$poles: \quad s = 0, \quad s = -\infty$$

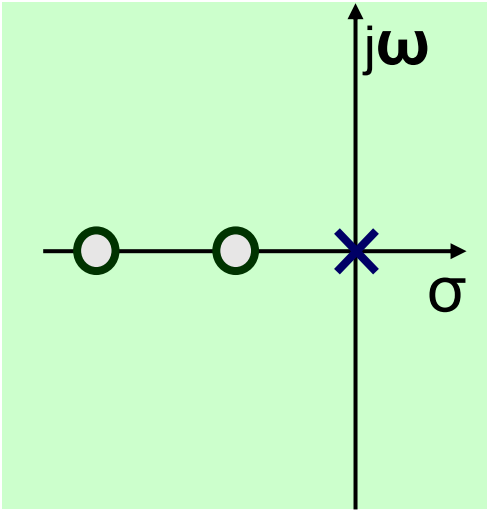
$$zeros: \quad s_{1,2} = \frac{-K_P \pm \sqrt{K_P^2 - 4K_I K_D}}{2K_D}$$

If $K_P > 2\sqrt{K_I K_D}$ $\Rightarrow$ real zeros

If $K_P < 2\sqrt{K_I K_D}$ $\Rightarrow$ complex conjugate zeros

Pole-zero configuration of a PID-controller (two possible configurations).





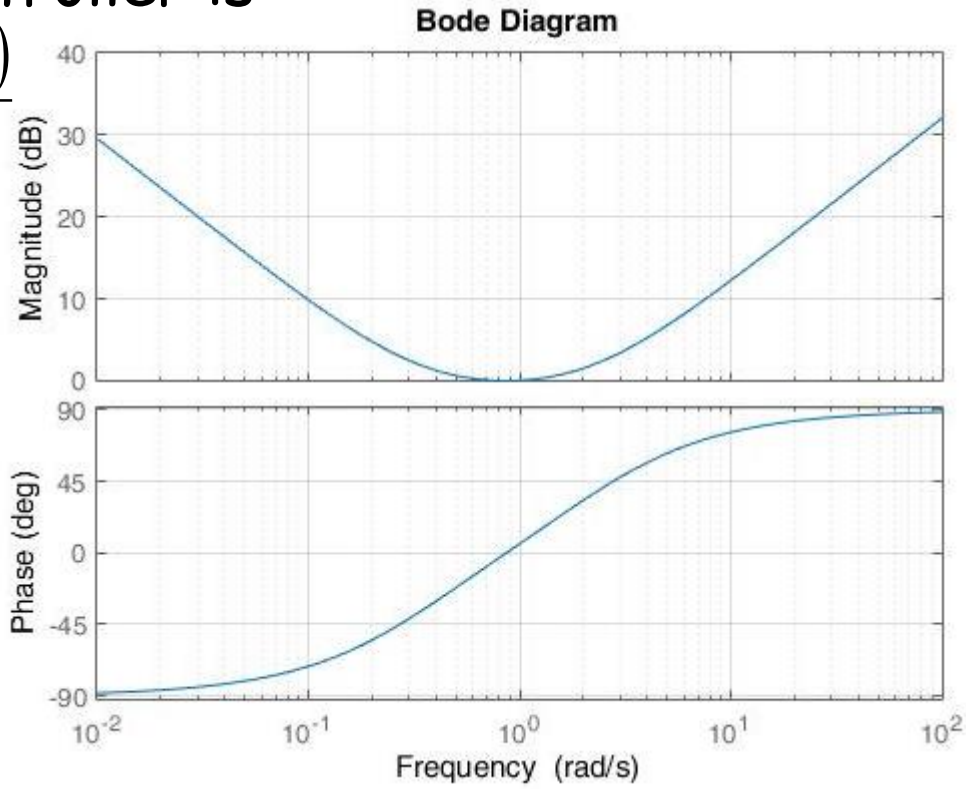
The zeros are always in the LHP.

PID-Controller

The sinusoidal TF of a PID controller is

$$G_c(j\omega) = \frac{K_I\left(1-\omega^2(K_D/K_I)+j\omega(K_P/K_I)\right)}{(j\omega)}$$

Its Bode diagram has the form



Bode diagram of a PID controller with $K_P = 1$, $K_I = 0.3$, and $K_D = 0.4$.

Note that a PID controller is a lead-lag controller and can be designed following the design procedure for them.

There are methods for tuning or determining the «best» PID controller parameters (given a system to be controlled).

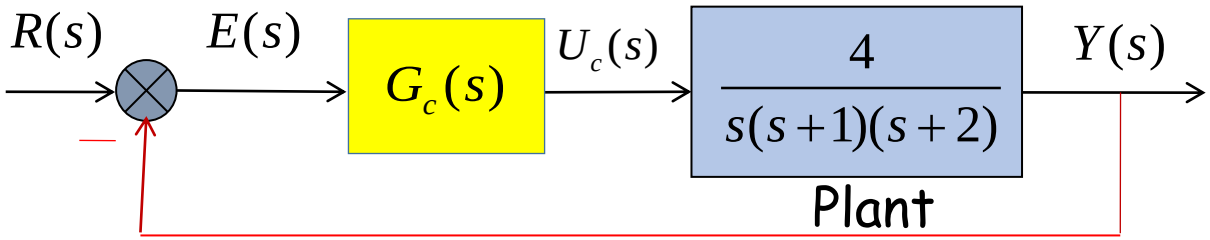
PID-Controller

A PID controller operates as follows:

- The integral term reduces the steady-state error to zero by taking the cumulative of the error.
- However, it decreases the speed of the response.
- The derivative term, on the other hand, «anticipates» the changes in the error by taking its derivative and provides an early action.
- This increases the speed of the response and decreases the overshoot.
- However, it may amplify high frequency noises and lead to large oscillations or even instability due to them.
- Large integral action may also lead to instability.
- Therefore, one should be careful while choosing the controller parameters.

PID-Controller

Example Consider the following control system



$$G(s)=\frac{4}{s(s+1)(s+2)}$$

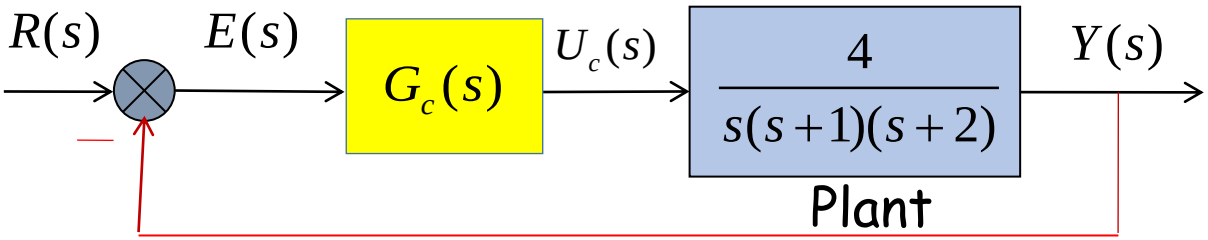
Design specifications:

It is required that the system has a phase margin of 50°, zero steady-state error to a ramp input, and 50% increase in the bandwidth.

Solution. Consider a PID controller

$$G_c(s)=\frac{K_D s^2+K_P s+K_I}{s}=\frac{K_D\left(s^2+(K_P/K_D)s+K_I/K_D\right)}{s}$$

PID-Controller



$$G_c(s) = \frac{K_D s^2 + K_P s + K_I}{s}$$

The steady-state error of the compensated CL system for a ramp input is

$$e_{ss} = 1/K_v$$

where $K_v = \lim_{s\to 0} sG_c(s)G(s) = \lim_{s\to 0} \frac{2K_I}{s} = \infty$ ⟹ $e_{ss} = 0$

Provided that the system is stable, the steady state error of the CL system to a ramp input is zero.

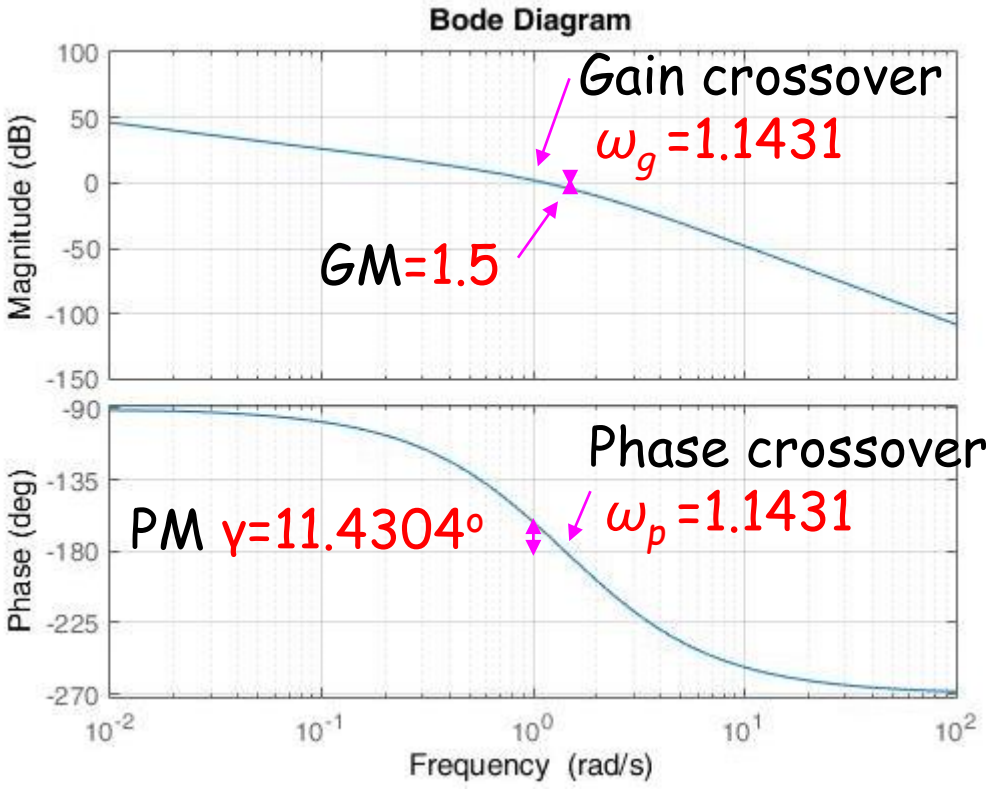
PID-Controller

The sinusoidal TF of the plant is

$$G(j\omega)=\frac{4}{(j\omega)(1+j\omega)(2+j\omega)}$$

$$=\frac{4}{-3\omega+j\omega(2-\omega^2)}$$

The Bode diagram of the uncompensated system is



The gain crossover frequency is at

$$\left|-3\omega+j\omega(2-\omega^2)\right|=4$$

$$\omega=1.1431 \text{ rad/sec}$$

The angle of $G(j\omega)$ at $\omega=\omega_g$ is

$$\measuredangle G(j1.1431)=-90^o-\tan^{-1}(1.1431)-\tan^{-1}(1.1431/2)=-168.5696^o$$

The phase margin is $\gamma=180^o-168.5696^o=11.4304^o$

The phase margin is unsatisfactory!

PID-Controller

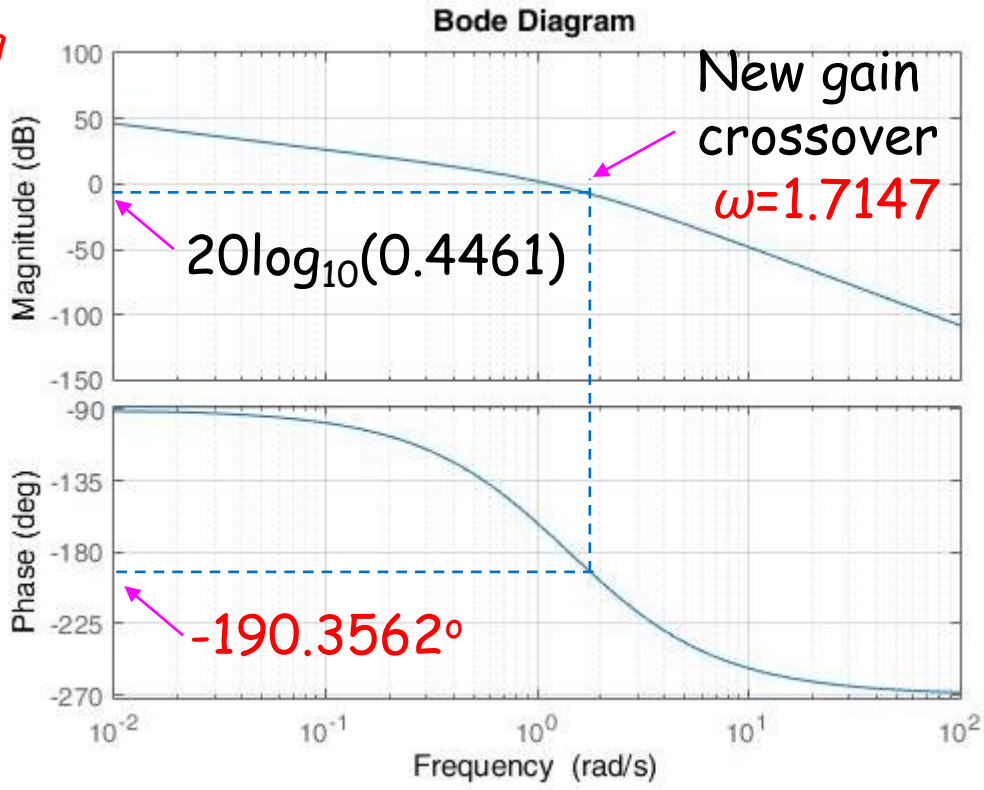
Choose the new ω_g after compensation to be 50% larger than the current ω_g

$$\omega = 1.7147 \text{ rad/sec}$$

$$|G(j1.7147)| = 0.4461$$

$$\angle G(j1.7147) = -190.3562^o$$

The Bode diagram of the uncompensated system is



Are the magnitude and phase angle of the plant at that frequency

To achieve a phase margin of 50° at ω=1.7147 rad/sec we need

$$\angle G_c(j1.7147) = -180^o + 190.3562^o + 50^o = 60.3562^o$$

Also, in order for ω=1.7147 rad/sec to be the new ω_g we need

$$|G_c(j1.7147)| = \frac{1}{0.4461} = 2.2416$$

PID-Controller

The magnitude/gain and the phase angle of a PID controller can be expressed as

$$|G_c(j\omega)| = \frac{\sqrt{\left(K_I - K_D\omega^2\right)^2 + \left(K_P\omega\right)^2}}{\omega}, \quad \measuredangle G_c(j\omega) = -90^o + \tan^{-1}\left(\frac{K_P\omega}{K_I - K_D\omega^2}\right)$$

For the requirements at ω=1.7147 rad/sec these become

$$\left(K_I - 2.9402K_D\right)^2 + 2.9402\left(K_P\right)^2 = 14.7738, \quad -90^o + \tan^{-1}\left(\frac{1.7147K_P}{K_I - 2.9402K_D}\right) = 60.3562^o$$

$$\left(K_I - 2.9402K_D\right)^2 + 2.9402\left(K_P\right)^2 = 14.7738, \quad \frac{1.7147K_P}{K_I - 2.9402K_D} = -0.5691$$

Here we have two equations with three unknowns.

We can choose one of the coefficients arbitrarily and determine the corresponding two other coefficients.

Choose the integral gain K_I=0.005 (small) and calculate the other two.

PID-Controller

Choose the integral gain K_I=0.005 (small), calculate the other two.

$$\left(0.005-2.9402K_D\right)^2+2.9402\left(K_P\right)^2=14.7738$$

$$\frac{1.7147K_P}{0.005-2.9402K_D}=-0.5691$$

$$K_P\cong 1.1087,\quad K_D\cong 1.1379$$

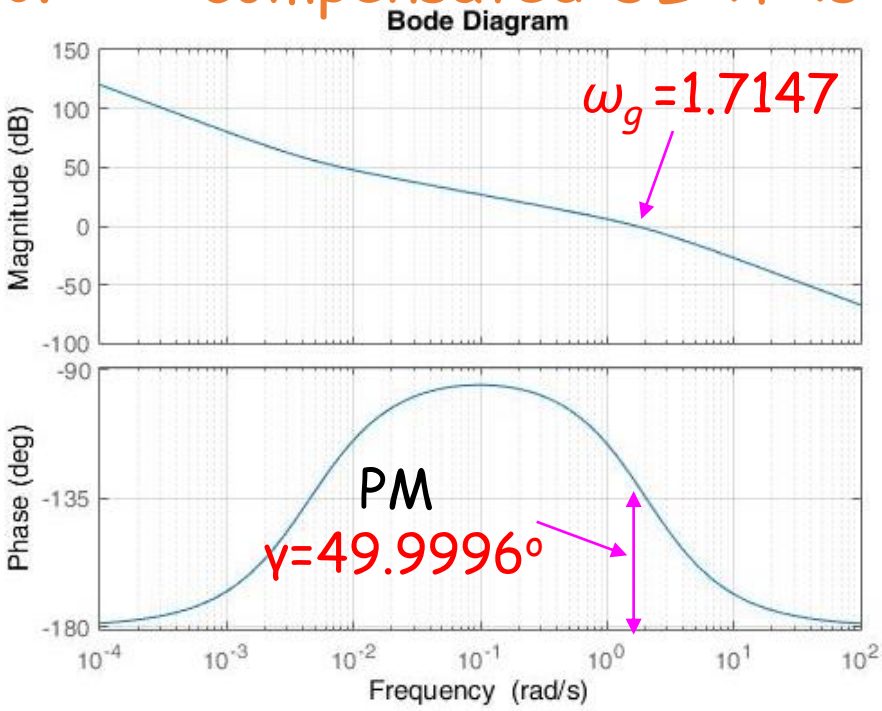
The PID controller TF is

$$G_c(s)=\left(1.1087+\frac{0.005}{s}+1.1379s\right)$$

The PID controller in time domain is

$$u_c(t)=1.1087e(t)+0.005\int_0^t e(\tau)d\tau+1.1379\dot{e}(t)$$

Bode diagram of the compensated OL TF is

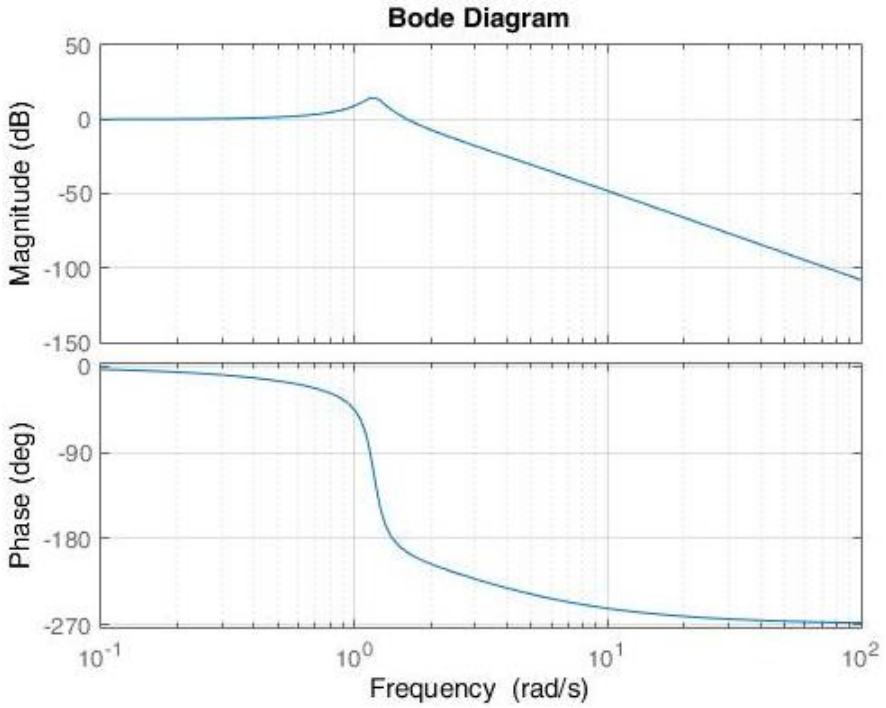


Compensated system has ∞ gain margin.

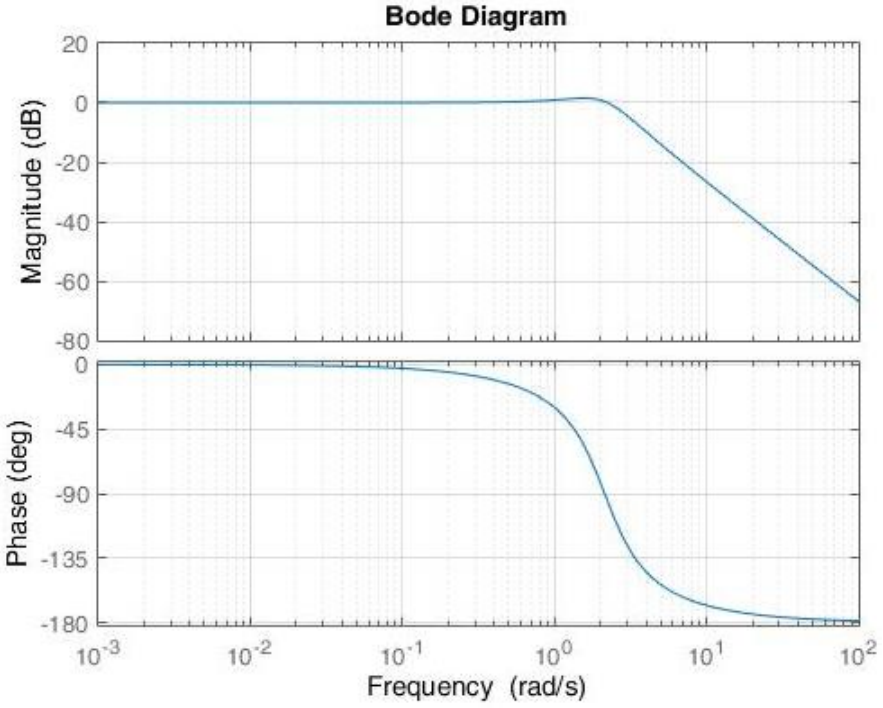
ω_g increased from 1.1431 rad/sec to 1.7147 rad/sec.

PID-Controller

Bode diagram of the uncompensated CL TF is



Bode diagram of the compensated CL TF is

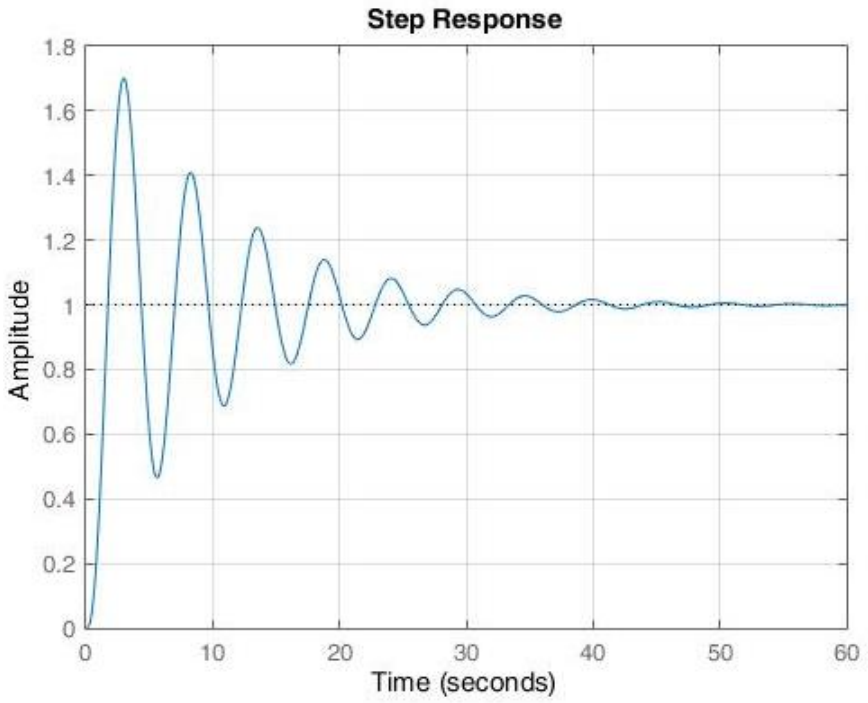


The resonant peak substantially decreased.

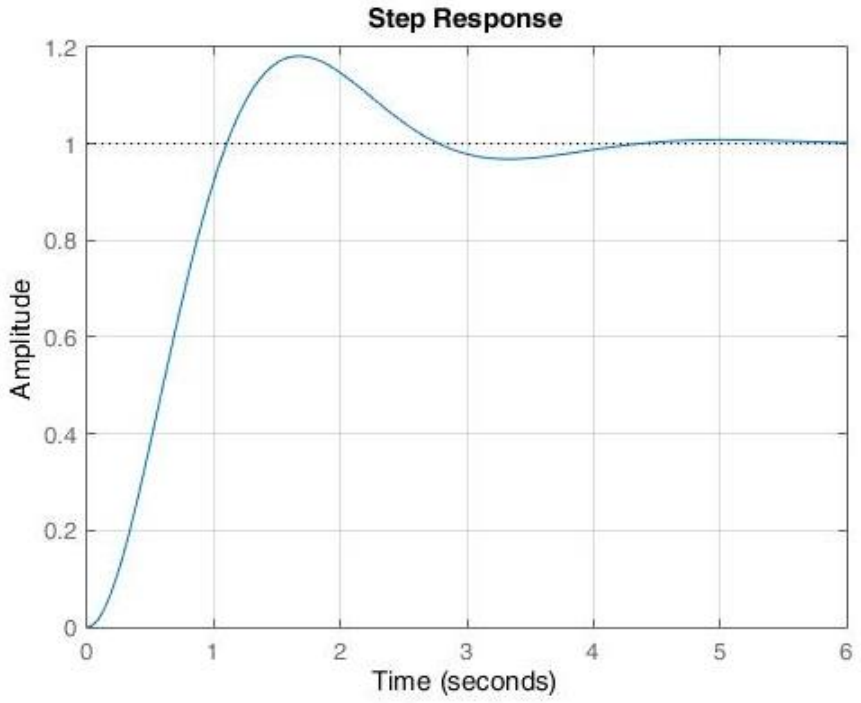
Bandwidth of the system increased from 1.7613 rad/sec to 2.7720 rad/sec (an increase of 57.38%).

PID-Controller

Step response of the uncompensated CL TF is



Step response of the compensated CL TF is

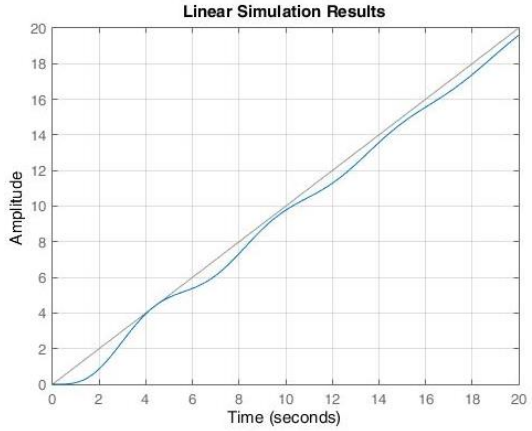


The maximum overshoot substantially decreased.

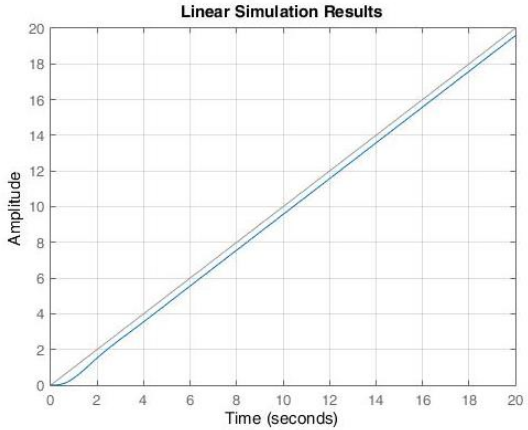
Oscillations of the response substantially decreased.

PID-Controller

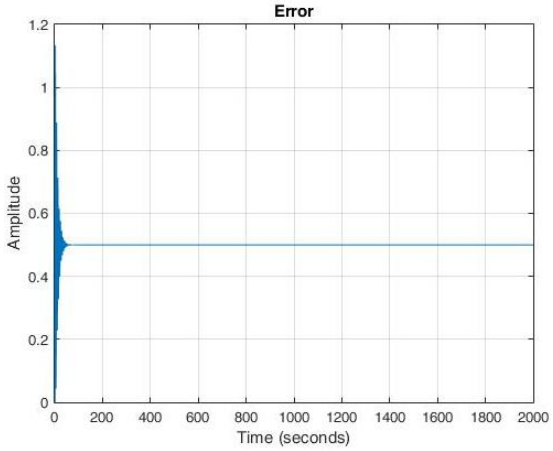
Uncompensated system



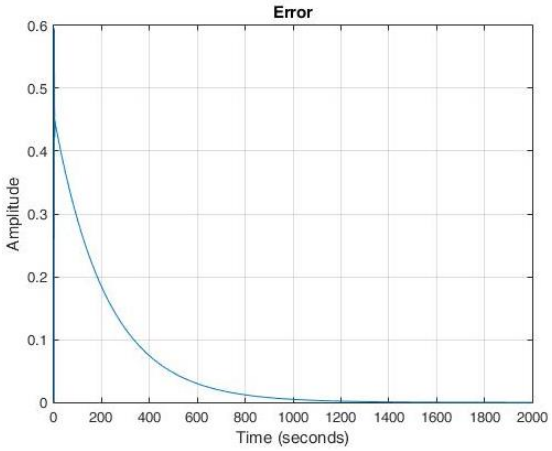
Compensated system



Ramp response



Error of the ramp response



Uncompensated system has constant steady state error.

Compensated system has zero steady state error.

Larger integral action would decrease the error faster.

Ziegler-Nichols Rules for Tuning PID Controllers

Ziegler-Nichols Rules for Tuning PID Controllers

$$G_c(s) = K_P + \frac{K_I}{s} + K_D s = K_P \left(1 + \frac{1}{T_I s} + T_D s \right)$$

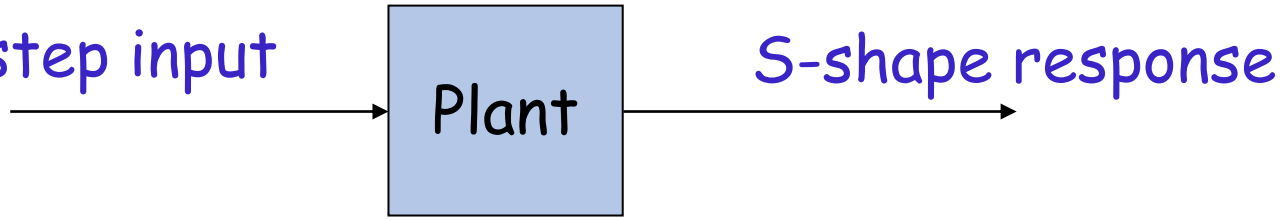
Ziegler and Nichols proposed rules for determining values of the proportional gain Kp, integral time Ti and derivative time Td based on the transient response of a given plant.

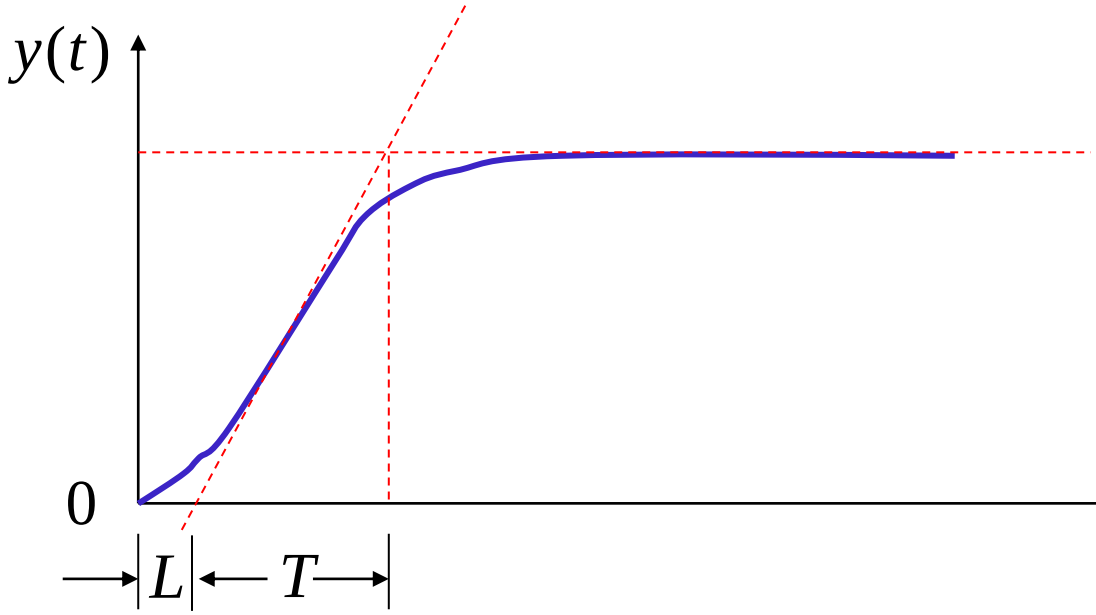
There are two methods called Ziegler-Nichols tuning rules: the first method and the second method.

First method

--Obtain the response of the plant to a unit-step input

--If the plant involves neither integrator nor dominant complex-conjugate poles, then its step response exhibits an S-shape curve , the first method can be applied.





The S-shape curve may be characterized by two constants: Delay time L and Time constant T

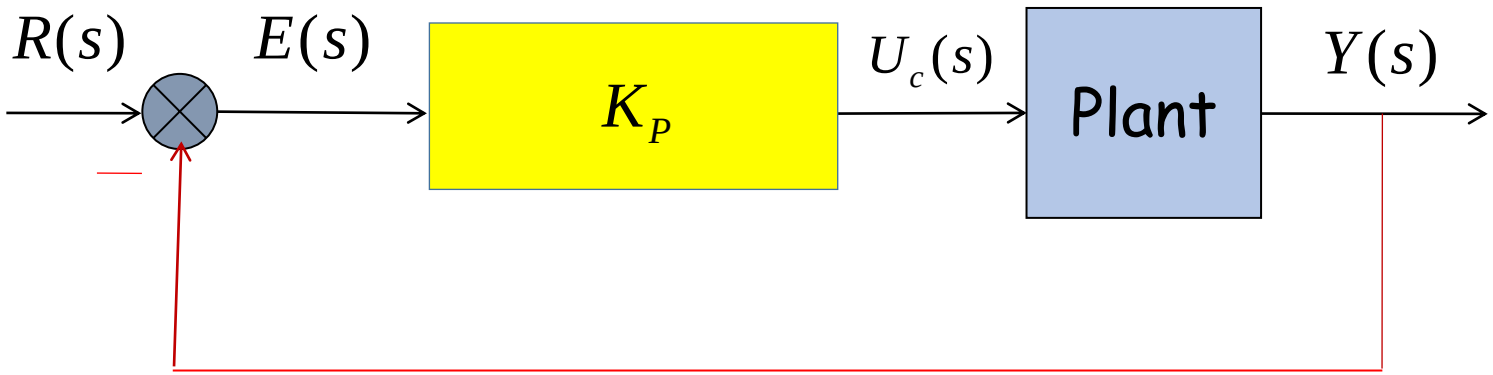
First method

Table I

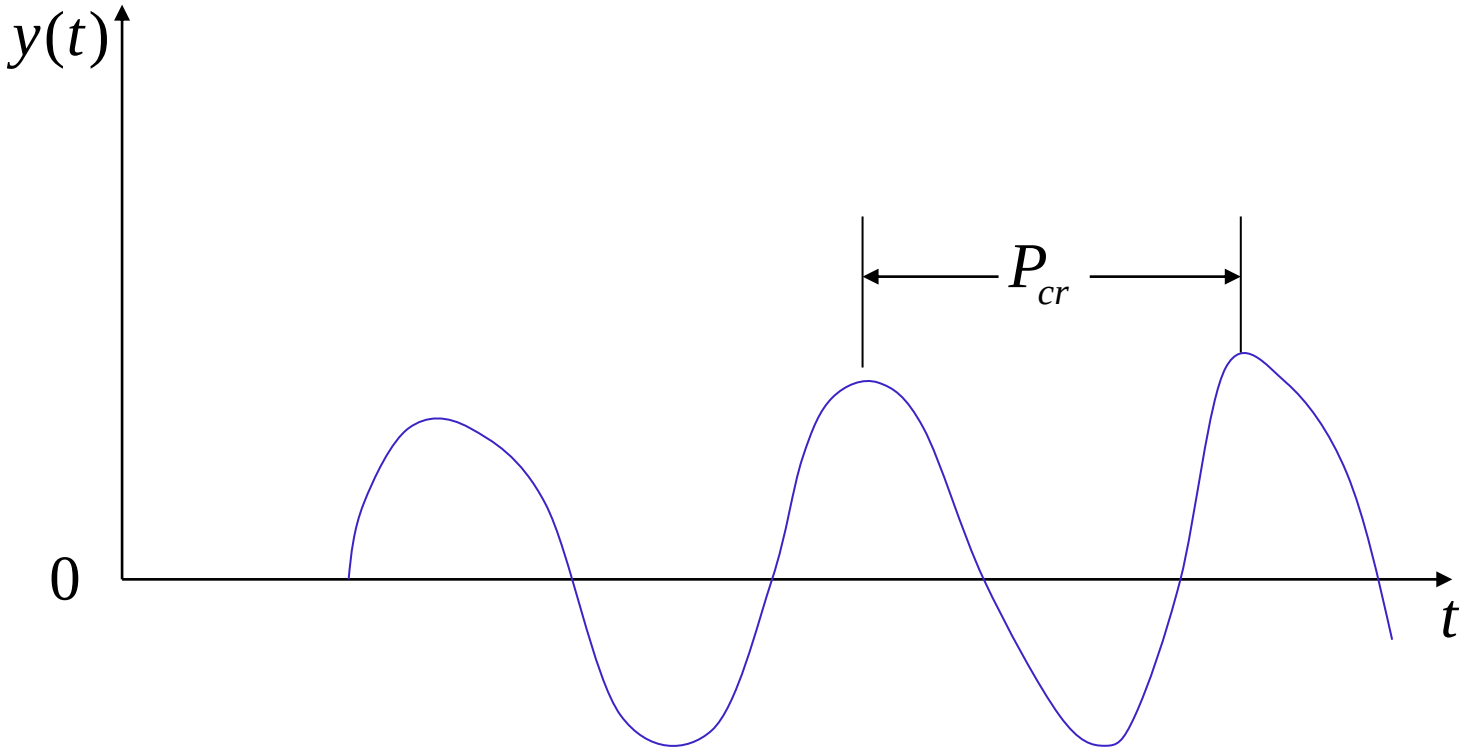
Type of Controller	K_P	T_I	T_D
P	$\frac{T}{L}$	∞	0
PI	$0.9\frac{T}{L}$	$\frac{L}{0.3}$	0
PID	$1.2\frac{T}{L}$	$2L$	$0.5L$

Second method

--Set $K_I = K_D = 0$ and use the proportional control only
--Increase K_P from 0 to ∞ to a critical value K_{cr} at which the output first exhibits sustained oscillations
--If the output does not exhibit sustained oscillations for whatever value K_P may take, then this method does not apply.



Second method



Second method

Table II

Type of Controller	K_P	T_I	T_D
P	$0.5K_{cr}$	∞	0
PI	$0.45K_{cr}$	$\frac{1}{1.2}P_{cr}$	0
PID	$0.6K_{cr}$	$0.5P_{cr}$	$0.125P_{cr}$

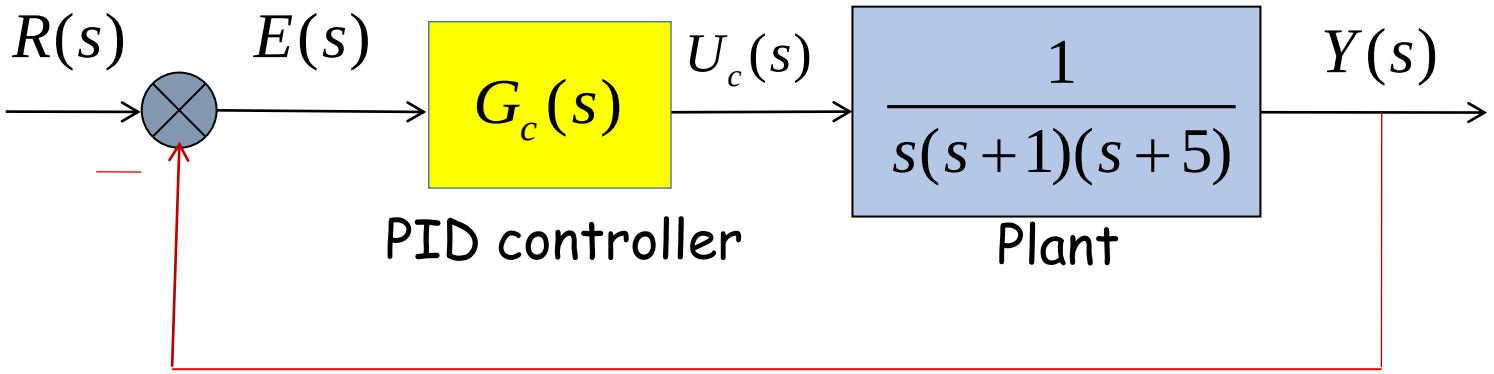
Comments

Ziegler-Nichols tuning rules have been widely used to tune PID controllers in process control systems where the plant dynamics are not precisely known.

If the plant dynamics are known, many analytical and graphical approaches for designing PID controllers are available, in addition to Ziegler-Nichols tuning rules

Example

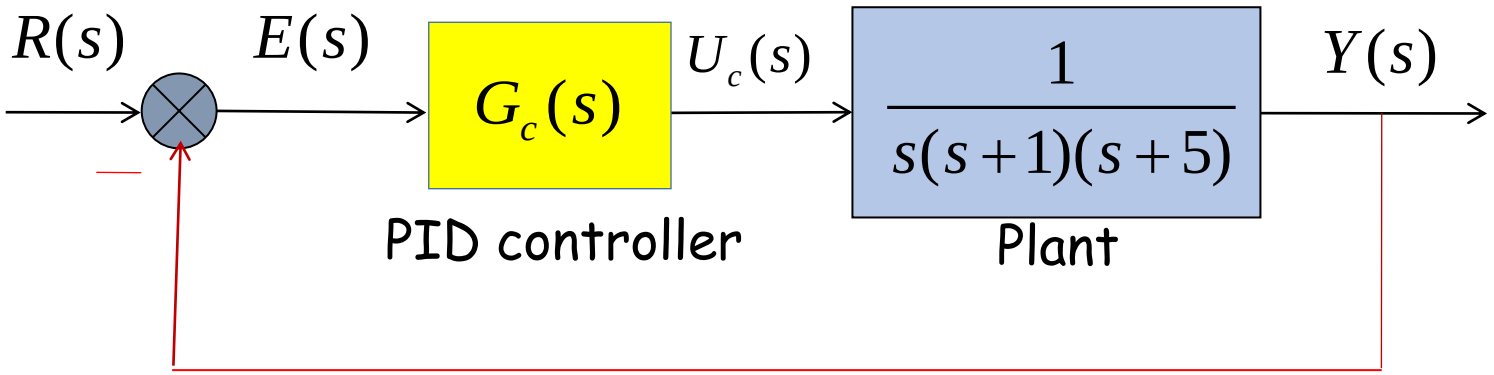
Consider the following control system



Design a PID controller to make the maximum overshoot of the system to be approximately 25% or less.

Solution. We start design the PID controller by applying Ziegler-Nichols rules.

Here the transfer function of the plant is known, we can use analytical method instead of experimental method.



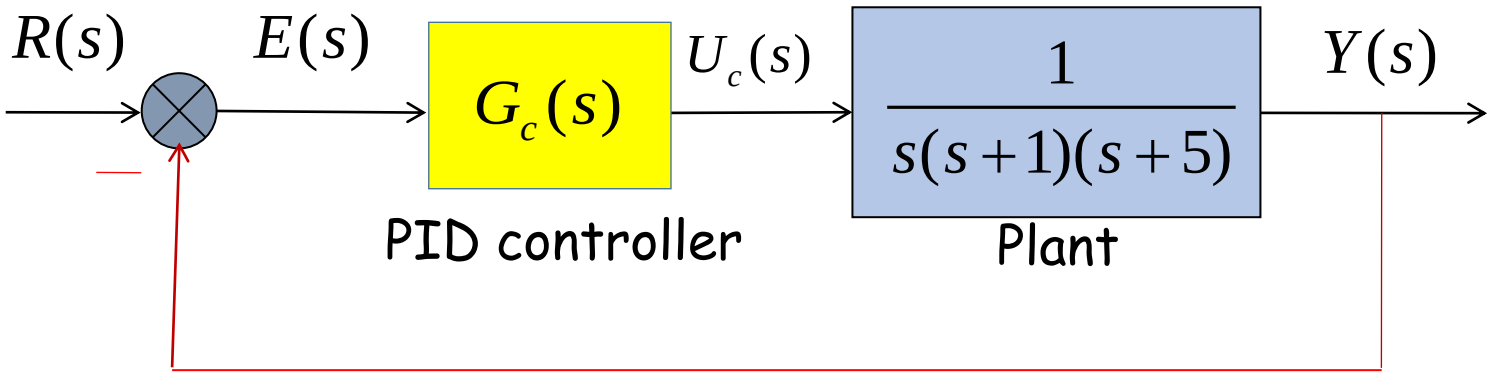
The PID controller has the transfer function

$$G_c(s) = K_P + \frac{K_I}{s} + K_D s = K_P \left(1 + \frac{1}{T_I s} + T_D s \right)$$

Since the plant has a integrator, we use the second method.

By setting $K_I = K_D = 0$, we obtain the closed-loop TF:

$$\frac{Y(s)}{R(s)} = \frac{K_P}{s(s+1)(s+5) + K_P}$$



The value of Kp that makes the system marginally stable so that sustained oscillation occurs can be obtained by use of Routh's stability criterion.

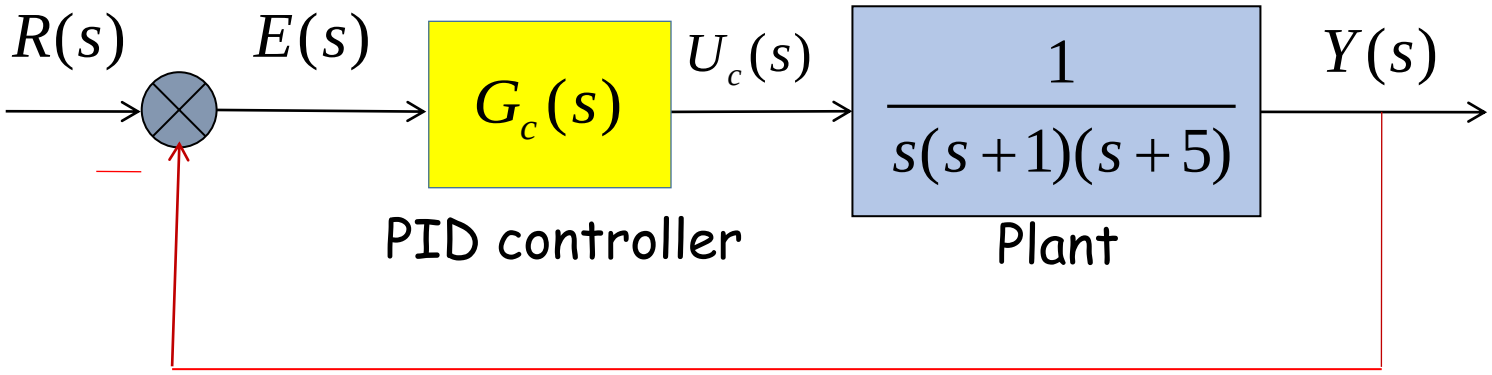
The CE of the closed-loop system is

$$s^3 + 6s^2 + 5s + K_P = 0$$

When Kp=30, the closed-loop system is marginally stable. Thus the critical gain $K_{cr} = 30$

The Routh's array

s^3	1	5
s^2	6	K_P
s^1	$\frac{30 - K_P}{6}$	
s^0	K_p	



With Kp set to Kcr(=30), the CE becomes

$$s^3 + 6s^2 + 5s + 30 = 0$$

To find the frequency of the sustained oscillation, we substitute s=jω into the CE as follows:

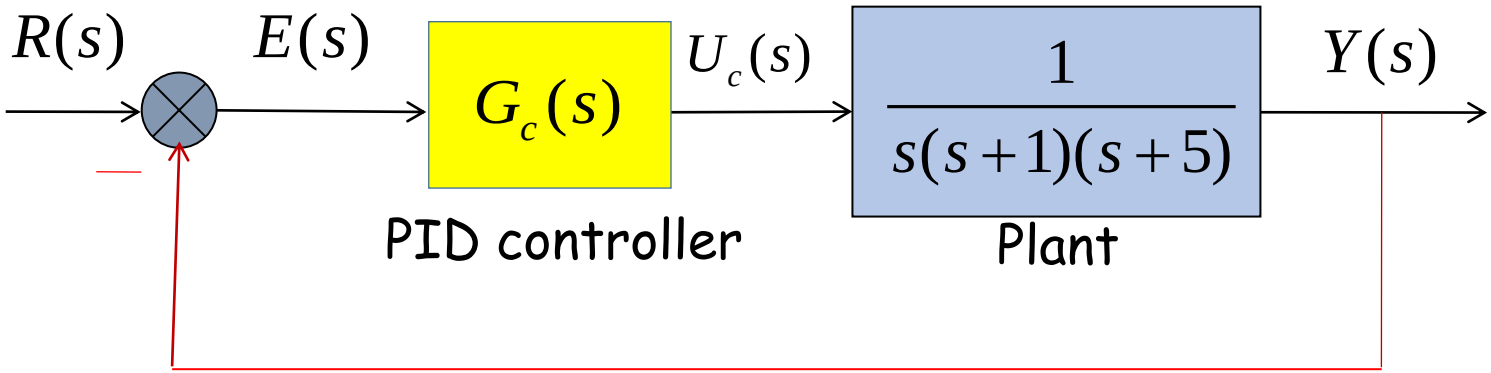
$$(j\omega)^3 + 6(j\omega)^2 + 5(j\omega) + 30 = 0$$

or

$$6(5-\omega^2) + j\omega(5-\omega^2) = 0 \quad \Rightarrow \quad \omega = \sqrt{5}$$

Hence the period of the sustained oscillation is

$$P_{cr} = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{5}} = 2.8099$$

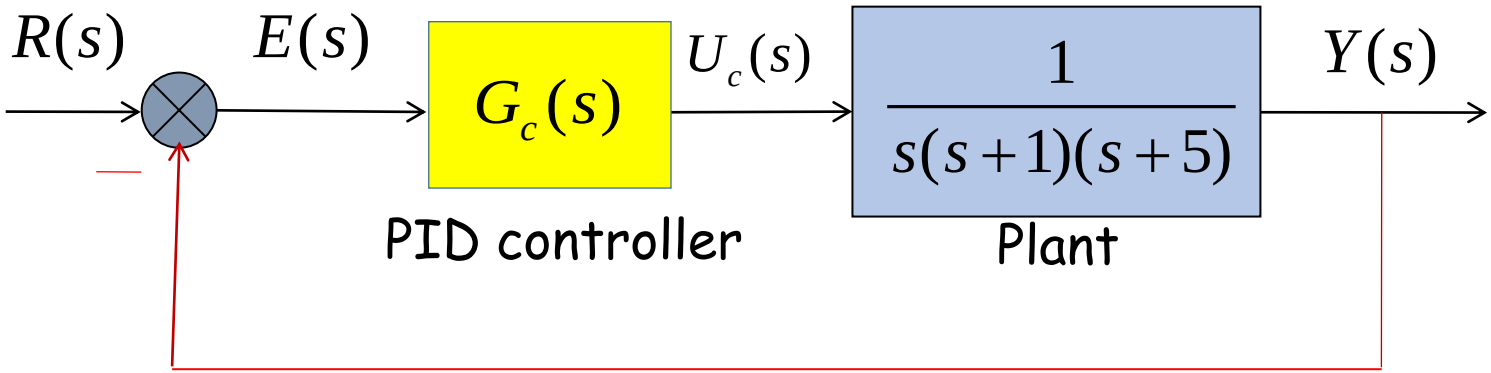


With Table II, we determine the parameters of the PID controller as follows

$$K_P = 0.6K_{cr} = 18$$
$$T_I = 0.5P_{cr} = 1.405$$
$$T_D = 0.125P_{cr} = 0.35124$$

The transfer function of the PID controller is thus

$$G_c(s) = 18\left(1 + \frac{1}{1.405s} + 0.35124s\right) = \frac{6.3223\left(s + 1.4235\right)^2}{s}$$

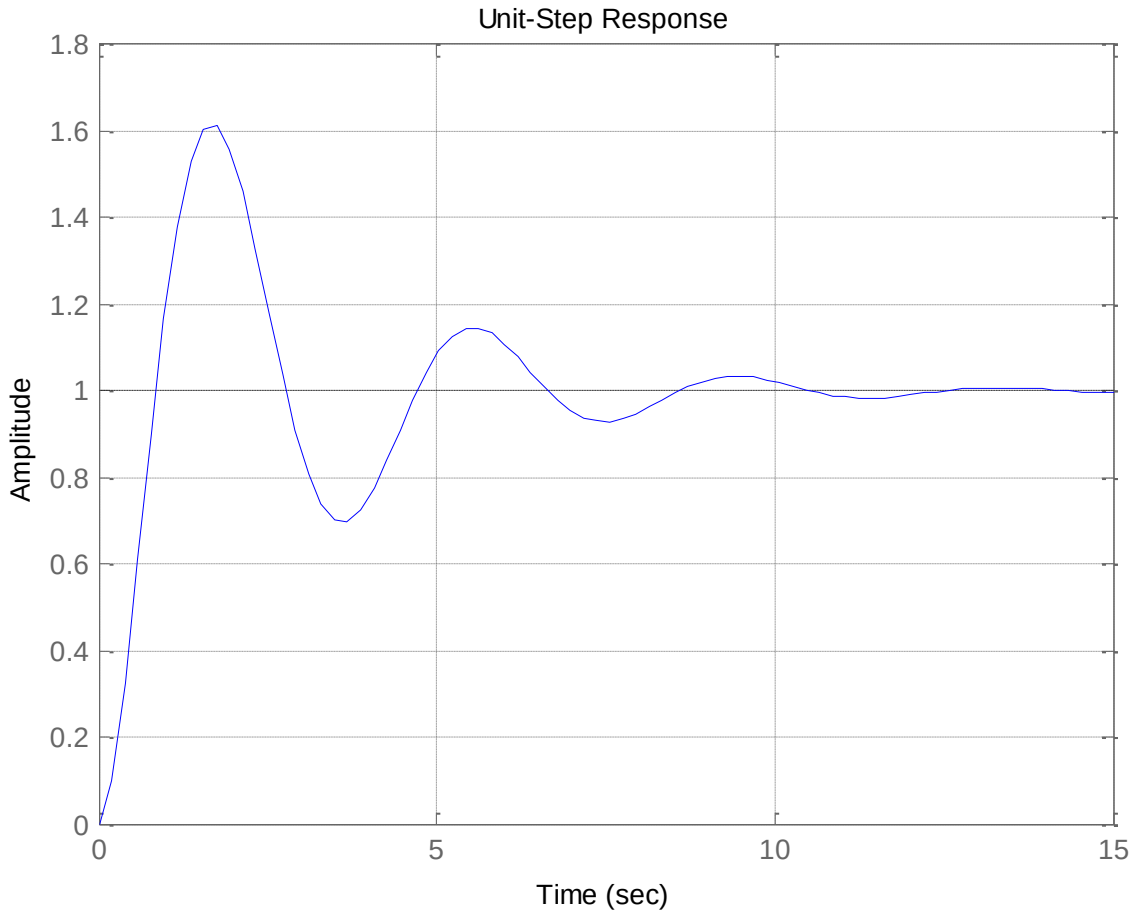


The closed-loop transfer function with the PID controller is

$$\frac{Y(s)}{R(s)} = \frac{6.3223s^2 + 18s + 12.811}{s^4 + 6s^3 + 11.3223s^2 + 18s + 12.811}$$

Now let us examine the unit-step response of the closed-loop system to see if it exhibits approximately 25% maximum overshoot.

```
>>num = [6.3223 18 12.811];
>>den = [1 6 11.3223 18 12.811];
>>step(num,den)
>>grid
```



The maximum overshoot is about 62% and is excessive with respect to the requirement of 25%.

The amount of maximum overshoot can be reduced by fine tuning the parameters of the PID controller.
Such fine tuning can be made on the computer.

$$G_c(s) = 18\left(1 + \frac{1}{1.405s} + 0.35124s\right) = \frac{6.3223(s+1.4235)^2}{s}$$

Move the zero locations

$$G_c(s) = 18\left(1 + \frac{1}{0.3077s} + 0.7692s\right) = \frac{13.846(s+0.65)^2}{s}$$

The maximum overshoot is reduced to 18%.

Increase the gain

$$G_c(s) = 39.42\left(1 + \frac{1}{0.3077s} + 0.7692s\right) = \frac{30.322(s+0.65)^2}{s}$$

The speed of response is increased. However, the maximum overshoot is also increased to 28%.

The parameters of the PID controller after fine tuning

$$K_P = 39.42$$
$$T_I = 3.077$$
$$T_D = 0.7692$$

Compared with the parameters obtained by Ziegler-Nichols' second method

$$K_P = 0.6K_{cr} = 18$$
$$T_I = 0.5P_{cr} = 1.405$$
$$T_D = 0.125P_{cr} = 0.35124$$

The new ones are approximately twice the values suggested by Ziegler-Nichols's method.

Note that the Ziegler-Nichols' tuning rule has provided a starting point for fine tuning.

Course Summary

What we have learned…

- How to build the mathematical model of a control system
  - Write differential equations based on physical laws
  - Linearize a nonlinear system about an operating point
  - Transform the differential equations to the s-domain by Laplace transform
  - Obtain the transfer function
  - Depict the control system with
    - Block diagram
    - Signal-flow graph (Mason's formula)

What we have learned…

- Time-domain analysis of control systems
  - Time-domain responses and time-domain specifications
    - Transient response (rise time, peak time, maximum overshoot, settling time)
    - Steady-state response (steady-state error)
  - Time-domain response of
    - First-order system
    - Prototype second-order system
  - Adding poles or zeros to loop or closed-loop transfer functions
    - Has varying effects on the transient response of the closed-loop system

What we have learned…

- Stability condition of a linear system
  - The roots of its characteristic equation (CE) must all be located in the left-half s-plane (LHP).
- Methods of determining stability
  - Routh-Hurwitz criterion
    - Test whether any of the roots of CE lie in RHP
    - Indicate the number of roots that lie on the jω-axis and in RHP
  - Integrability of the impulse response
    - The impulse response must be absolutely integrable

What we have learned…

- Methods of determining stability
  - Nyquist criterion
    - Semi-graphical method -- Nyquist plot
    - Analyze closed-loop stability based on the loop transfer function $G(s)H(s)$
    - Closed-loop stability criterion: $N=-P$
    - For minimum-phase loop function: $N=0$
  - Bode diagram
    - Plot the magnitue of the loop transfer function $G(j\omega)H(j\omega)$ in dB and the phase in degrees versus frequency ω
    - Closed-loop stability can be determined by observing the behavior of the plots (gain margin, phase margin)

What we have learned…

- Control system design technique
  - Root-locus design
    - A graphical method
    - Investigate how the roots of the CE of a LTI system move when one or more parameters vary
  - Frequency-response design
    - A graphical method
    - Provide information different from what we get from root-locus analysis
    - Can be applied to high-order system
    - Can use the data from the measurements of a physical system without deriving its mathematical model

What we have learned…

- To change the performance of a control system
  - Vary the gain K
  - Add poles
  - Add zeros
  - Design a Compensator
    - Lead Compensator
    - Lag Compensator
    - Lead-Lag Compensator
  - Design a PID Controller
    - PI Controller
    - PD Controller
    - PID Controller

END OF THE COURSE